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MSML606

TSP Proposal

Team Composition - solo

AI Usage Statement - The only AI used within this project was the Google AI agent to confirm my recollection of the nearest neighbors algorithm. Other than that, no AI was used within this proposal.

Dataset:

In this project, I will be using the SimpleMaps US Cities Dataset (simplemaps.com/data/us-cities). This dataset is free to use for the public and contains geographic information for over 30,000 cities. For this project specifically, the program will choose a random sample of 12-15 cities to represent the traveling salesperson problem. The only fields that will be necessary are the city name and state, and the coordinates. By combining these 2 fields, we can create a fully connected weighted graph represented by a distance matrix.

Problem and Algorithm:

Problem Statement -

The traveling salesperson problem is a classic problem within graph theory. It involves trying to find the best path to take that will visit all cities in the sample, and returning to the starting city. The main objective is to find the shortest amount of time/distance this would take, given that you can visit the cities in any order.

Algorithm -

To solve this problem, I will be using an algorithm called nearest neighbor (NN). This is a greedy approach with a time complexity of $O(n^2)$. The core idea is to repeatedly visit the closest unvisited city until all cities have been visited.

- 1.) Start with a random city and mark it visited
- 2.) Visit the nearest unvisited city (from the current one) and go to it
- 3.) Repeat step 2 until all cities have been visited
- 4.) Add the starting node to the end of the path (from wherever the last node was) and return